MRNet Knee MRI Classification

Transfer Learning with Multi-Plane Fusion using ResNet18

CECS 553 — Machine Vision

Problem

Knee MRI diagnosis is bottlenecked by cost, speed, and human inconsistency.

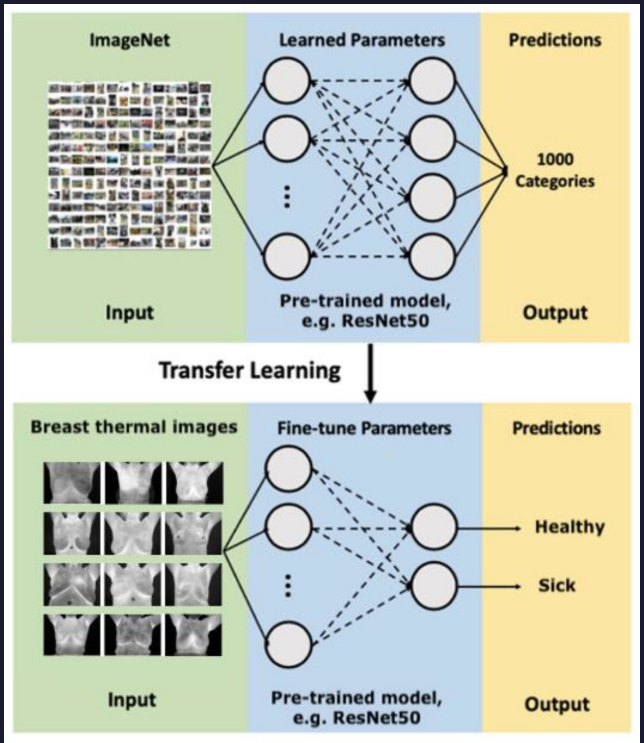
- Knee injuries are one of the most common reasons for MRI referrals
- *Accurate interpretation requires specialized radiologists*
- *Growing exam volume outpaces available experts*

Proposed Solution

Apply transfer learning and fine-tuning to automate multi-condition knee MRI classification.

High Level Goals

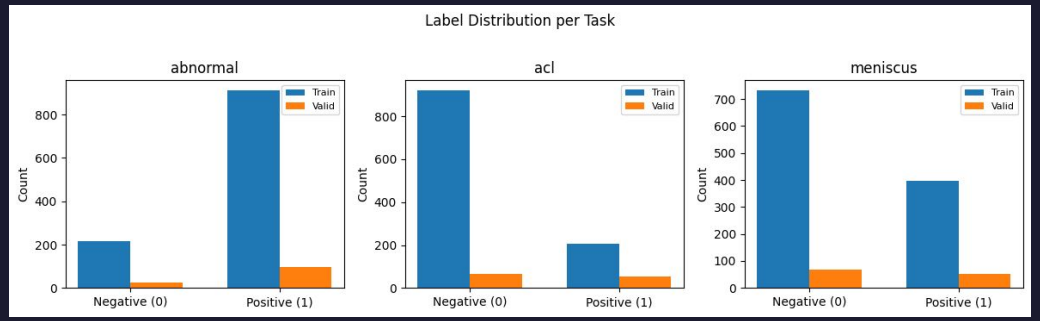
- *Reduce reliance on specialists*
- *Speeds up diagnosis*
- *Improves consistency*
- *Catches subtle findings*



Dataset Overview: MRNet

1,370 knee MRI exams with binary labels for three clinical conditions.

Task	Train Positive	Train Negative	% Positive	Valid Positive	Valid Negative
Abnormal	913	217	80.8%	95	25
ACL	208	922	18.4%	54	66
Meniscus	397	733	35.1%	52	68



Key observations:

- Abnormal is heavily imbalanced
- ACL is rare (18.4%)

Stanford AIMI

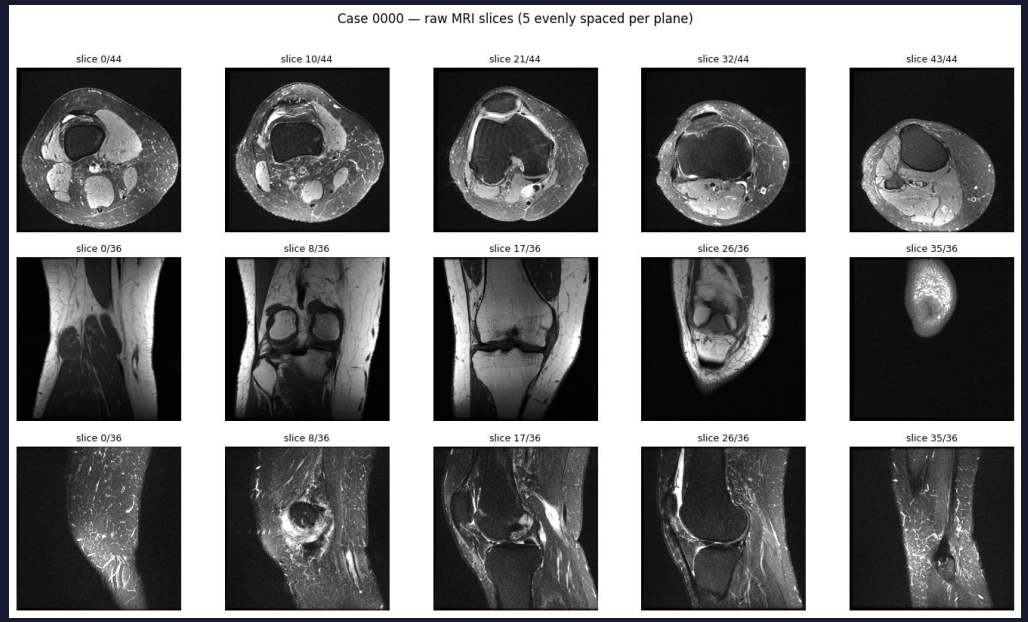
Center for Artificial Intelligence in Medicine and Imaging

Understanding a Knee MRI Exam

Each exam captures the knee from three viewing angels and produces three separate sets of images slices per angel

- ***Axial**: Horizontal cross-sections*
- ***Coronal**: Front-to-back cross-sections*
- ***Sagittal**: Side cross-sections*

- *A stack of 25–40 grayscale 256x256 images per plane*
- *Each image is a single cross-section at a specific depth*
- *Brighter pixels represent denser tissue; darker areas represent fluid or space*



Challenges

Building a classifier for knee MRI introduces both modeling and data-specific hurdles.

Modeling Challenges

- *Limited labeled data — Only 1,130 training exams*
- *Domain gap — Pretrained models learn from natural images (ImageNet)*
- *Multi-view reasoning — Radiologists cross-reference all three planes together*
- *Variable-length volumes — Each patient has a different number of slices per plane*

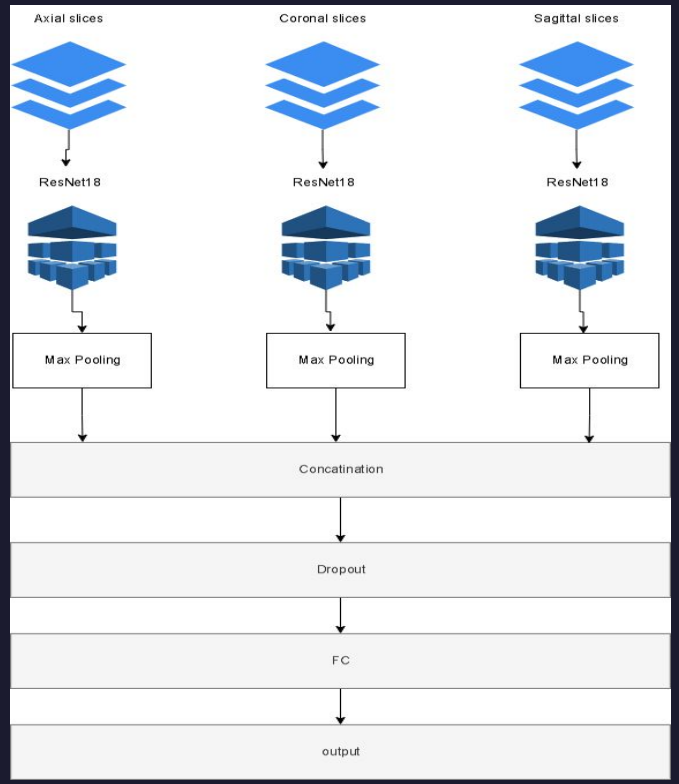
Data Challenges

- *Class imbalance — Abnormal dominates at 80.8% positive, while ACL is only 18.4%*
- *Grayscale single-channel input — MRI slices are single-channel, but ImageNet backbones expect 3-channel RGB*
- *No standardized slice count — Volume lengths vary per patient and plane*

Solution: Multi-Plane Fusion Architecture

Three pretrained backbones extract features per plane; max-pool collapses slices; fusion head classifies.

- All slices per plane → ResNet18 → (S, 512) per-slice features
- Max-pool over slices → (1, 512) per plane
- Concatenate 3 planes → (1, 1536) fused representation
- Linear(1536, 1) → probability of condition



Training Strategy

Freeze the backbone first to stabilize the classification head, then unfreeze to adapt to the MRI domain

Phase 1 – Feature Extraction

- *All three ResNet18 backbones frozen — weights don't change*
- *Only the classification head trains (~1,537 parameters)*
- *Head learns to map pretrained ImageNet features to knee pathology predictions*

Phase 2 – Fine-tuning

- *All layers unfrozen — full end-to-end training (~33M parameters)*
- *Fresh Adam optimizer created with same learning rate*
- *Backbone adapts its filters from natural image patterns to MRI tissue textures*

Hyperparameters and Design Decisions

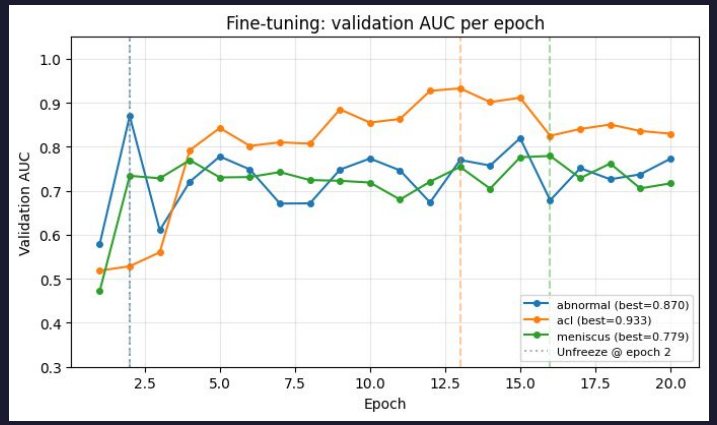
Hyperparameter	Value
Backbone	ResNet18
Optimizer	Adam
Learning Rate	1e-4
Batch Size	1
Loss Function	BCEWithLogitsLoss
Evaluation Metric	ROC, AUC
Epochs	20
Freeze Epochs	1
Dropout	0.0

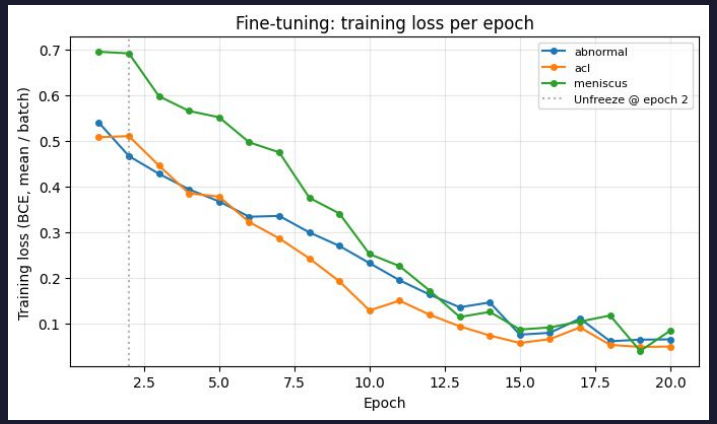
Hyperparameter	Value
Weight Decay	0.0
Input Size	224x224
Slice Sampling	None

Data Preprocessing Pipeline

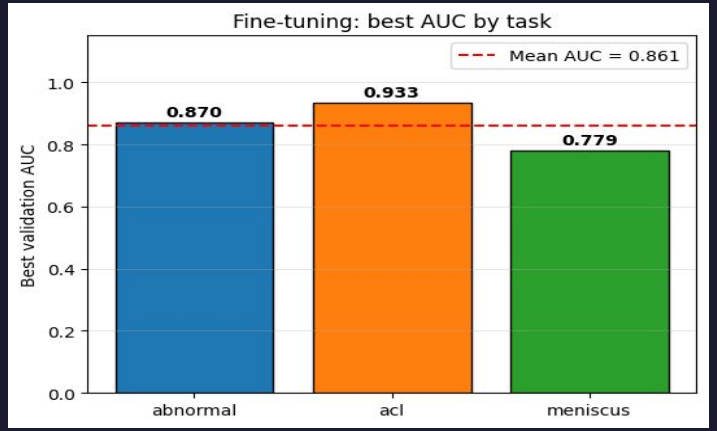
- *Load .npy → (S, 256, 256) grayscale slices*
- *Normalize → min-max scale each volume to [0, 1]*
- *Resize → 256x256 down to 224x224 for ImageNet compatibility*
- *Replicate → 1 channel to 3 channels (grayscale to RGB format)*
- *Augment (training only) → random horizontal flip at 50% probability*

Results: Three-Task Training





Average AUC: 0.861

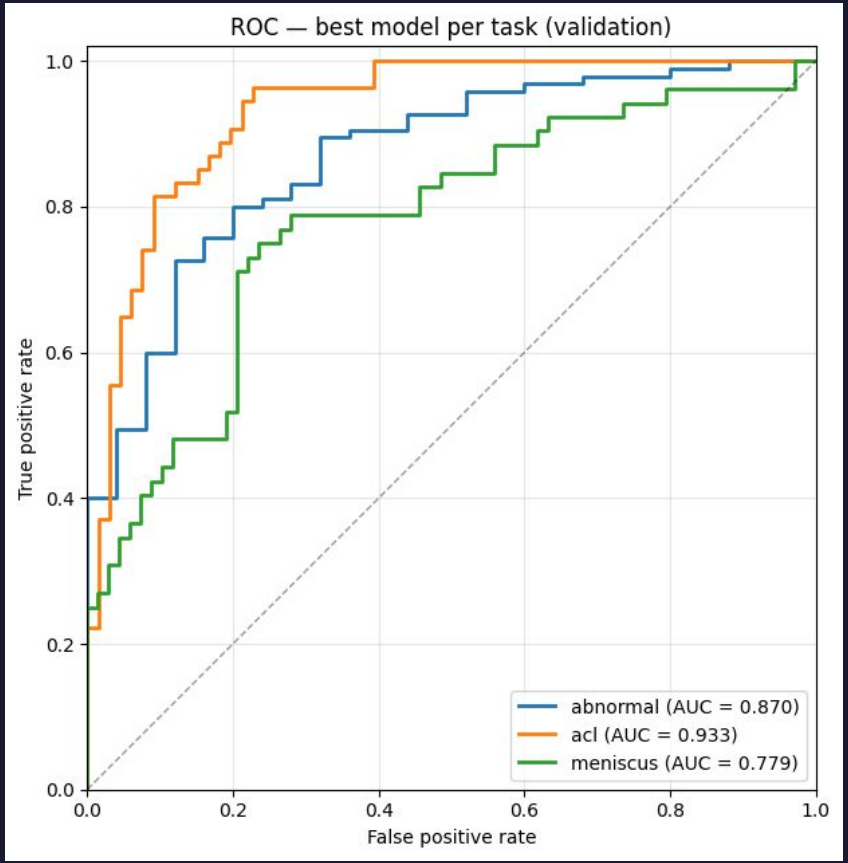


Settings: 20 epochs, lr=1e-4, batch_size=1, all slices, CUDA (RTX 3080)

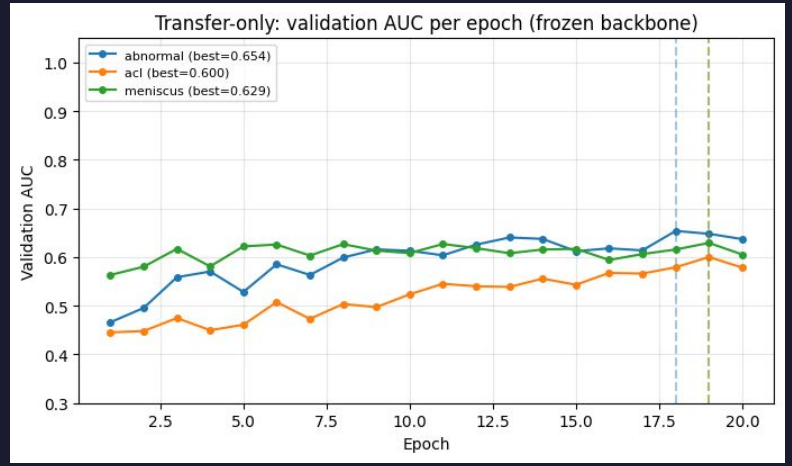
Total training time: 348.9 minutes (~5.8 hours) for all 3 tasks

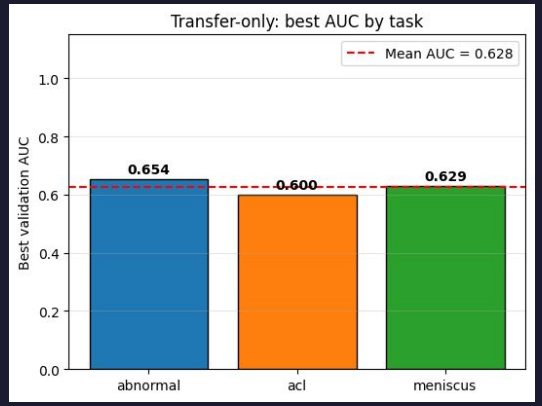
Best checkpoint saved per task when validation AUC improved

Results: ROC Curve

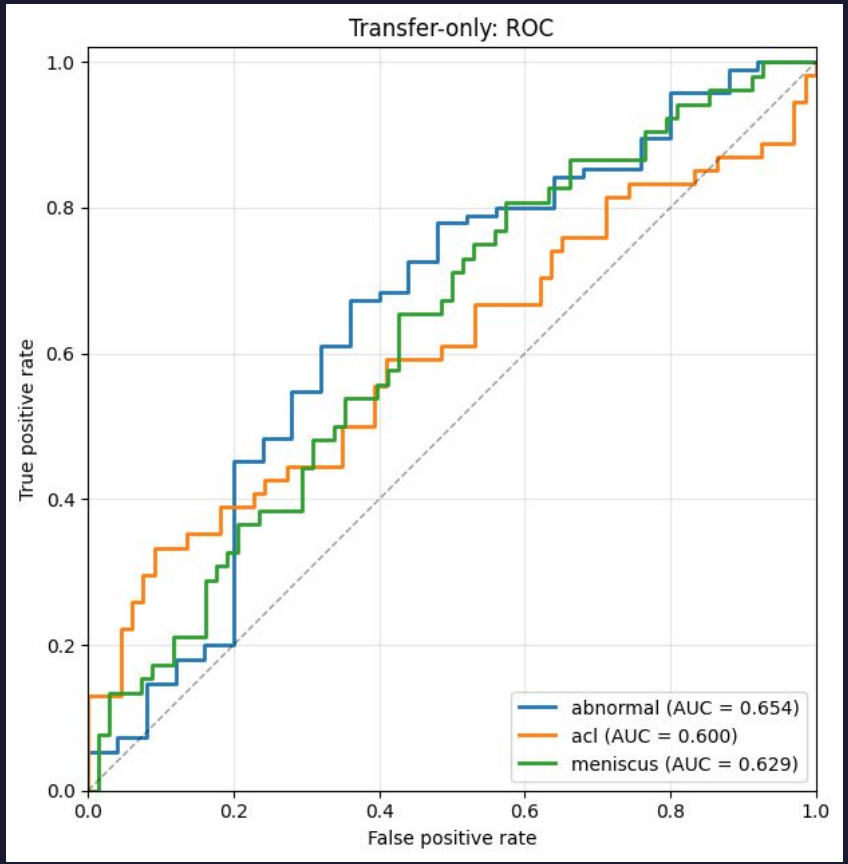


Results: Transfer-Only | Three Task Training





Results: Transfer-Only | ROC Curve



Key Takeaways

- Fine-tuning wins on all 3 tasks (validation AUC vs transfer-only).
- Mean AUC: ~0.86 (fine-tune) vs ~0.63 (classification head only).
- By task: abnormal 0.87 vs 0.65 · ACL 0.93 vs 0.60 · meniscus 0.78 vs 0.63.
- Biggest gap: ACL (+0.33). Smallest: meniscus (+0.15).

References

1. Bien, N., et al. "Deep-learning-assisted diagnosis for knee magnetic resonance imaging: Development and retrospective validation of MRNet." PLOS Medicine, 2018. DOI: 10.1371/journal.pmed.1002699

2. Stanford AIMI. "MRNet: Knee MRI's." DOI: 10.71718/rcbp-8c35 https://aimi.stanford.edu/datasets/mrnet-knee-mris

3. He, K., et al. "Deep Residual Learning for Image Recognition." CVPR, 2016.

4. Deng, J., et al. "ImageNet: A Large-Scale Hierarchical Image Database." CVPR, 2009.